

NAKUL KUMBRIA

Software Development Engineer / Backend Engineer

Oceanside, CA | (714) 924-0481 | nakul.kumbria.uni@outlook.com | [linkedin.com/in/nakul-kumbria-a5b343212](https://www.linkedin.com/in/nakul-kumbria-a5b343212) | github.com/nakicool007

PROFESSIONAL EXPERIENCE

Software Engineer / IT Project Manager | Thermo Fisher Scientific, Carlsbad, CA Nov 2024 – Present

- Engineered an AI-powered chatbot using Python, OpenAI API, LangChain, and Streamlit to answer natural-language questions across internal datasets, replacing manual knowledge lookup.
- Developing an AI-powered project insights portal for EPPM, integrating backend data through OData to generate organization-specific summaries and PowerPoint decks for leadership reviews.
- Implemented Retrieval-Augmented Generation (RAG) pipelines in Databricks to ground chatbot answers in internal datasets, replacing manual document search.
- Built and maintained cloud-native applications using AWS EC2, S3, and Cloud9 with scalable, modular architectures.
- Automated sprint reporting from Jira using Python and REST APIs, extracting Agile metrics and delivering scheduled analytics emails to stakeholders each sprint.
- Analyzed EC2 usage patterns and identified underutilized resources for downsizing or shutdown to reduce AWS spend.
- Administer Databricks non-production environments, managing clusters, access control, data pipelines, and workspace reliability.

Systems Analyst (Software & Data Engineering Focus) | Thermo Fisher Scientific, Carlsbad, CA Jun 2022 – Nov 2024

- Built and optimized ETL pipelines using AWS S3, Databricks, Redshift, and SQL, feeding financial and sales analytics.
- Engineered Python automation scripts to ingest data from SharePoint and Salesforce into SQL Server, reducing manual data handling.
- Created dashboards using Power BI, Tableau, and Salesforce to support executive-level financial decision-making.
- Automated recurring manual workflows using UiPath, Power Automate, and Selenium.
- Managed and queried relational databases using SQL Server, DBeaver, and MySQL.

Web Developer Intern | Experis IT (Manpower Group), Gurugram Jun 2021 – Aug 2021

- Developed PHP (CodeIgniter) backend components and MySQL schemas for enterprise web applications.

Web Developer Intern | SUNLOC, Delhi Dec 2020 – May 2021

- Built full-stack PHP/MySQL applications and implemented database backup and recovery mechanisms.

PROJECTS

Art Collection Explorer | Python, Node.js, Express, Docker, Google Cloud Run, REST APIs Apr 2026 – May 2026

- Developed and deployed a cloud-hosted full-stack artwork discovery platform integrating the Harvard Art Museums and Metropolitan Museum of Art APIs.
- Built backend API aggregation services supporting filtering, pagination, caching, and asynchronous external API requests.
- Designed frontend features including dynamic search, favorites management, and real-time result rendering.

Movie Explorer | Python, Flask, Docker, NGINX, Google Cloud Run Mar 2026 – Apr 2026

- Engineered and deployed a full-stack movie discovery platform using Flask and REST APIs to aggregate movie metadata, trailers, ratings, and reviews.
- Designed modular backend services for API orchestration, request handling, and secure configuration management across external data sources.
- Built frontend workflows with dynamic routing, movie search, and detailed content views.

Power Plant Output Prediction | Python, scikit-learn, pandas Sep 2026

- Compared linear, polynomial, interaction, and KNN regression models on 9,568 observations to predict power plant energy output.
- Evaluated normalization, 100 KNN neighborhood sizes, statistical tests, and backward elimination to identify the best-performing approach.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, PHP, JavaScript

Frontend, Backend & Frameworks: Spring Boot, Node.js, Express, Flask, Laravel, GraphQL, React, HTML, CSS, Bootstrap

Databases: MySQL, SQL Server, MongoDB, Redshift, Postgres

Cloud & Data: AWS (EC2, S3, Cloud9, SageMaker), Google Cloud Run, Docker, Databricks, PySpark

Tools: Git, Selenium, Power BI, Tableau, Jira, Salesforce

Concepts: Data Structures & Algorithms, REST APIs, Cloud Architecture, Automation, System Design

EDUCATION

University of Southern California Jan 2025 – Present

Master of Science in Computer Science

University of Texas Permian Basin Jan 2018 – Dec 2022

Bachelor of Science in Computer Science | GPA: 3.851 | Dean's List | President's Honor Roll